The DMAIC Sprint Board

Every Excel statistics function and tool, pinned to the DMAIC phase where a Green or Black Belt reaches for it.
Exam traps: STDEV.S is overall σ (Ppk, not Cpk) · Z.TEST is one-tailed · use T.INV.2T for ± critical t · I-chart limits use 2.66 × MR̄, not 3 × STDEV.S

SPRINT RULE
Work top to bottom.
No test before a baseline.

D — DEFINE

Size the problem and find the vital few

13 TOOLS

COUNT THE DEFECTS — 5

COUNT(value1, [value2], ...) — The sample size for every other formula

COUNTA(value1, [value2], ...) — Checking for text entries in numeric columns

COUNTBLANK(range) — Missing-data audit

COUNTIF(range, criteria) — Defects beyond a spec limit

COUNTIFS(criteria_range1, criteria1, ...) — Contingency tables and DPU by category

STRATIFY AND PARETO — 8

GROUPBY(row_fields, values, function, [field_headers], ...) HELPER — Stratify by heat, shift, line or supplier

PIVOTBY(row_fields, col_fields, values, function, ...) HELPER — Side-by-side comparisons such as LPA vs TWA

PERCENTOF(data_subset, data_all) HELPER — Pareto-style contributions

FILTER(array, include, [if_empty]) HELPER — Slicing a subgroup without copying

UNIQUE(array, [by_col], [exactly_once]) HELPER — Listing categories before grouping

SORT(array, [sort_index], [sort_order], [by_col]) HELPER — Ordering data for plots and ranks

SUBTOTAL(function_num, ref1, [ref2], ...) HELPER — Summaries that follow a filter

AGGREGATE(function_num, options, ref1, [ref2], ...) HELPER — Robust summaries on messy data

M — MEASURE

Baseline the process and its capability

82 TOOLS

DESCRIBE THE DATA — 31

AVERAGE(number1, [number2], ...) — Default centre of a data set

AVERAGEA(value1, [value2], ...) — Only when logical values must count

AVERAGEIF(range, criteria, [average_range]) — Mean for one shift, grade or line

AVERAGEIFS(average_range, criteria_range1, criteria1, ...) — Mean by heat and orientation together

MEDIAN(number1, [number2], ...) — Skewed data such as delays and cycle times

MODE.SNGL(number1, [number2], ...) — Discrete readings with repeats

MODE.MULT(number1, [number2], ...) — Data with two or more peaks

GEOMEAN(number1, [number2], ...) — Growth rates, ratios, log-normal data

HARMEAN(number1, [number2], ...) — Averaging rates such as units per hour

TRIMMEAN(array, percent) — A robust mean when outliers are present

STDEV.S(number1, [number2], ...) — Almost every quality calculation on sample data

STDEV.P(number1, [number2], ...) — Only when you hold the whole population

STDEVA(value1, [value2], ...) — Rarely; text becomes 0 and distorts results

STDEVPA(value1, [value2], ...) — Rarely; same caution as STDEVA

VAR.S(number1, [number2], ...) — ANOVA, F-tests and pooled variance

VAR.P(number1, [number2], ...) — Complete populations only

VARA(value1, [value2], ...) — Rarely

VARPA(value1, [value2], ...) — Rarely

AVEDEV(number1, [number2], ...) — A spread summary that resists outliers

DEVSQ(number1, [number2], ...) — Building ANOVA sums of squares by hand

SKEW(number1, [number2], ...) — Quick normality and symmetry screen

SKEW.P(number1, [number2], ...) — Skewness of a complete population

KURT(number1, [number2], ...) — Screening for heavy or light tails

MAX(number1, [number2], ...) — Range and worst-case checks

MAXA(value1, [value2], ...) — Rarely

MAXIFS(max_range, criteria_range1, criteria1, ...) — Worst reading per grade or line

MIN(number1, [number2], ...) — Range and minimum-spec checks

MINA(value1, [value2], ...) — Rarely

MINIFS(min_range, criteria_range1, criteria1, ...) — Lowest toughness per heat

LARGE(array, k) — Top-k values and order statistics

SMALL(array, k) — Bottom-k values and order statistics

TURN DATA INTO ODDS — 41

NORM.DIST(x, mean, standard_dev, cumulative) — Share of output below a limit, ppm out of spec

NORM.INV(probability, mean, standard_dev) — Percentile limits; simulation with RAND()

NORM.S.DIST(z, cumulative) — p-values for z statistics

NORM.S.INV(probability) — Critical z, sample size, sigma level

GAUSS(z) — Quick area from the centre

PHI(x) — Plotting the bell curve

T.DIST(x, deg_freedom, cumulative) — Lower-tailed t tests

T.DIST.RT(x, deg_freedom) — Upper-tailed t tests

T.DIST.2T(x, deg_freedom) — Two-sided p-value for a t statistic

T.INV(probability, deg_freedom) — One-sided critical values; negative for small p

T.INV.2T(probability, deg_freedom) — Confidence intervals and two-sided tests

CHISQ.DIST(x, deg_freedom, cumulative) — Lower tail of a variance test

CHISQ.DIST.RT(x, deg_freedom) — p-value for chi-square tests with any df

CHISQ.INV(probability, deg_freedom) — Upper limit of a variance confidence interval

CHISQ.INV.RT(probability, deg_freedom) — Chi-square table lookups; variance intervals

F.DIST(x, deg_freedom1, deg_freedom2, cumulative) — Lower tail of a variance-ratio test

F.DIST.RT(x, deg_freedom1, deg_freedom2) — ANOVA and regression p-values

F.INV(probability, deg_freedom1, deg_freedom2) — Lower limit of a variance-ratio interval

F.INV.RT(probability, deg_freedom1, deg_freedom2) — F tables for ANOVA

BINOM.DIST(number_s, trials, probability_s, cumulative) — Defectives in a sample, pass or fail data

BINOM.DIST.RANGE(trials, probability_s, number_s, [number_s2]) — Acceptance-sampling ranges in one step

BINOM.INV(trials, probability_s, alpha) — Acceptance numbers and exact critical counts

NEGBINOM.DIST(number_f, number_s, probability_s, cumulative) — Inspections until a set number of finds

HYPGEOM.DIST(sample_s, number_sample, population_s, number_pop, cumulative) — Sampling small lots without replacement

POISSON.DIST(x, mean, cumulative) — Defects per unit, c- and u-chart probabilities

EXPON.DIST(x, lambda, cumulative) — Time between failures, constant failure rate

WEIBULL.DIST(x, alpha, beta, cumulative) — Life data and wear-out reliability

LOGNORM.DIST(x, mean, standard_dev, cumulative) — Repair times, particle sizes, skewed data

LOGNORM.INV(probability, mean, standard_dev) — Percentile limits for skewed data

GAMMA.DIST(x, alpha, beta, cumulative) — Waiting times for several events

GAMMA.INV(probability, alpha, beta) — Percentiles of gamma-shaped data

GAMMA(number) — Building other distributions by hand

GAMMALN(x) — Large factorials without overflow

GAMMALN.PRECISE(x) — Same use, newer algorithm

BETA.DIST(x, alpha, beta, cumulative, [A], [B]) — PERT task times, proportions

BETA.INV(probability, alpha, beta, [A], [B]) — Exact binomial confidence limits

PERMUT(number, number_chosen) — Counting sequences and orders

PERMUTATIONA(number, number_chosen) — Codes and passwords style counting

PROB(x_range, prob_range, [lower_limit], [upper_limit]) — Custom discrete distributions

COMBIN(number, number_chosen) HELPER — Counting samples where order does not matter

COMBINA(number, number_chosen) HELPER — Counting with replacement

POSITION AND SHAPE — 10

FREQUENCY(data_array, bins_array) — Histograms and goodness-of-fit counts

QUARTILE.EXC(array, quart) — Box plots that match Minitab

QUARTILE.INC(array, quart) — Matching older Excel workbooks

PERCENTILE.EXC(array, k) — P5 or P95 limits that match Minitab

PERCENTILE.INC(array, k) — Percentiles that include 0 and 1

PERCENTRANK.EXC(array, x, [significance]) — Where one value sits in the data

PERCENTRANK.INC(array, x, [significance]) — Same, inclusive method

RANK.AVG(number, ref, [order]) — Normal probability plots and rank tests

RANK.EQ(number, ref, [order]) — League tables and Pareto ordering

STANDARDIZE(x, mean, standard_dev) — Flagging points beyond three standard deviations

A — ANALYZE

Prove which Xs drive Y

39 TOOLS

PUT A ± ON IT — 2

CONFIDENCE.T(alpha, standard_dev, size) — Confidence interval for a mean, sigma unknown

CONFIDENCE.NORM(alpha, standard_dev, size) — Confidence interval for a mean, sigma known

TEST THE DIFFERENCE — 4

T.TEST(array1, array2, tails, type) — Comparing two means

Z.TEST(array, x, [sigma]) — Mean against a target when sigma is known

F.TEST(array1, array2) — Checking variances before choosing T.TEST type

CHISQ.TEST(actual_range, expected_range) — Counts by category; contingency tables

MEASURE THE LINK — 7

CORREL(array1, array2) — Strength and direction of a straight-line link

PEARSON(array1, array2) — Legacy sheets; identical result

RSQ(known_y's, known_x's) — Share of variation explained by one x

COVARIANCE.S(array1, array2) — Building r by hand on sample data

COVARIANCE.P(array1, array2) — Matches the ToolPak Covariance tool

FISHER(x) — Confidence interval on r; comparing two r values

FISHERINV(y) — Second half of the r interval

MODEL Y = F(X) — 7

LINEST(known_y's, [known_x's], [const], [stats]) — Simple or multiple regression with inference

SLOPE(known_y's, known_x's) — Change in y per unit of x

INTERCEPT(known_y's, known_x's) — y when x is zero

STEYX(known_y's, known_x's) — Width of prediction intervals

TREND(known_y's, [known_x's], [new_x's], [const]) — Fitted values, residuals, predictions

LOGEST(known_y's, [known_x's], [const], [stats]) — Growth or decay curves

GROWTH(known_y's, [known_x's], [new_x's], [const]) — Predictions from an exponential model

FULL OUTPUT TABLES (TOOLPAK) — 19

Anova: Single Factor · Data › Data Analysis › Anova: Single Factor — Three or more group means

Anova: Two-Factor With Replication · Data › Data Analysis › Anova: Two-Factor With Replication — Two factors with repeats

Anova: Two-Factor Without Replication · Data › Data Analysis › Anova: Two-Factor Without Replication — Randomized block designs

t-Test: Paired Two Sample for Means · Data › Data Analysis › t-Test: Paired Two Sample for Means — Before and after on the same units

t-Test: Two-Sample Assuming Equal Variances · Data › Data Analysis › t-Test: Two-Sample Assuming Equal Variances — Independent groups, equal spread

t-Test: Two-Sample Assuming Unequal Variances · Data › Data Analysis › t-Test: Two-Sample Assuming Unequal Variances — Independent groups, unequal spread

z-Test: Two Sample for Means · Data › Data Analysis › z-Test: Two Sample for Means — Two means with known variances

F-Test Two-Sample for Variances · Data › Data Analysis › F-Test Two-Sample for Variances — Comparing two variances

Descriptive Statistics · Data › Data Analysis › Descriptive Statistics — One-click summary

Regression · Data › Data Analysis › Regression — Full regression output

Correlation · Data › Data Analysis › Correlation — Several variables at once

Covariance · Data › Data Analysis › Covariance — Several variables at once

Histogram · Data › Data Analysis › Histogram — Quick frequency table

Rank and Percentile · Data › Data Analysis › Rank and Percentile — Relative standing table

Random Number Generation · Data › Data Analysis › Random Number Generation — Static simulated data

Sampling · Data › Data Analysis › Sampling — Drawing an audit sample

Moving Average · Data › Data Analysis › Moving Average — Trend smoothing

Exponential Smoothing · Data › Data Analysis › Exponential Smoothing — Level-only forecasts

Fourier Analysis · Data › Data Analysis › Fourier Analysis — Vibration and periodic signals

I — IMPROVE

Find, simulate and prove the fix

18 TOOLS

OPTIMISE AND SIMULATE — 18

Goal Seek · Data › What-If Analysis › Goal Seek — Which sigma gives Cpk 1.33?

Data Table (one variable) · Data › What-If Analysis › Data Table (column input cell) — Sensitivity of cost or yield to one input

Data Table (two variables) · Data › What-If Analysis › Data Table (row and column input cells) — Cpk map across mean and sigma

Scenario Manager · Data › What-If Analysis › Scenario Manager — Best, likely and worst cases

Solver (Simplex LP) · Data › Solver › Select a Solving Method: Simplex LP — Blends, product mix, scheduling

Solver (GRG Nonlinear) · Data › Solver › Select a Solving Method: GRG Nonlinear — Response optimization, curve fitting

Solver (Evolutionary) · Data › Solver › Select a Solving Method: Evolutionary — Models with IF, LOOKUP or rounding

Forecast Sheet · Data › Forecast › Forecast Sheet — One-click seasonal forecast

RAND() HELPER — Driving simulations through .INV functions

RANDARRAY([rows], [columns], [min], [max], [integer]) HELPER — Thousands of draws in one formula

RANDBETWEEN(bottom, top) HELPER — Random sampling of rows

SEQUENCE(rows, [columns], [start], [step]) HELPER — Input grids for sensitivity tables

LET(name1, name_value1, calculation_or_name2, ...) HELPER — Readable one-cell simulations

LAMBDA([parameter1, ...], calculation) HELPER — Custom test statistics and models

MAP(array1, lambda_or_array) HELPER — Formula-based data tables that recalculate fast

SUMPRODUCT(array1, [array2], ...) HELPER — Objective and constraint cells for Solver

MMULT(array1, array2) HELPER — Regression by matrices; linear models

MINVERSE(array) HELPER — Normal equations for regression by hand

C — CONTROL

Hold the gain and keep the workbook clean

58 TOOLS

WATCH FOR DRIFT — 6

FORECAST.LINEAR(x, known_y's, known_x's) — Straight-line trend forecasts

FORECAST(x, known_y's, known_x's) — Older workbooks only

FORECAST.ETS(target_date, values, timeline, [seasonality], [data_completion], [aggregation]) — Time series with trend and seasonality

FORECAST.ETS.CONFINT(target_date, values, timeline, [confidence_level], [seasonality], [data_completion], [aggregation]) — Error bands around ETS forecasts

FORECAST.ETS.SEASONALITY(values, timeline, [data_completion], [aggregation]) — Confirming the cycle before trusting a forecast

FORECAST.ETS.STAT(values, timeline, statistic_type, [seasonality], [data_completion], [aggregation]) — Judging forecast accuracy

REPLACE LEGACY NAMES — 39

BETADIST → BETA.DIST
BETAINV → BETA.INV
BINOMDIST → BINOM.DIST
CHIDIST → CHISQ.DIST.RT
CHIINV → CHISQ.INV.RT
CHITEST → CHISQ.TEST
CONFIDENCE → CONFIDENCE.NORM
COVAR → COVARIANCE.P
CRITBINOM → BINOM.INV
EXPONDIST → EXPON.DIST
FDIST → F.DIST.RT
FINV → F.INV.RT
FORECAST → FORECAST.LINEAR
FTEST → F.TEST
GAMMADIST → GAMMA.DIST
GAMMAINV → GAMMA.INV
HYPGEOMDIST → HYPGEOM.DIST
LOGINV → LOGNORM.INV
LOGNORMDIST → LOGNORM.DIST
MODE → MODE.SNGL
NEGBINOMDIST → NEGBINOM.DIST
NORMDIST → NORM.DIST
NORMINV → NORM.INV
NORMSDIST → NORM.S.DIST
NORMSINV → NORM.S.INV
PERCENTILE → PERCENTILE.INC
PERCENTRANK → PERCENTRANK.INC
POISSON → POISSON.DIST
QUARTILE → QUARTILE.INC
RANK → RANK.EQ
STDEV → STDEV.S
STDEVP → STDEV.P
TDIST → T.DIST.RT or T.DIST.2T
TINV → T.INV.2T
TTEST → T.TEST
VAR → VAR.S
VARP → VAR.P
WEIBULL → WEIBULL.DIST
ZTEST → Z.TEST

HAND OFF TO MINITAB OR PYTHON — 12

Normality test (Anderson-Darling, Shapiro-Wilk)
In Excel: Normal probability plot with NORM.S.INV and RANK.AVG; check SKEW and KURT

One- and two-proportion tests
In Excel: z statistic with NORM.S.DIST; exact test with BINOM.DIST

Nonparametric tests (sign, Wilcoxon, Mann-Whitney, Kruskal-Wallis)
In Excel: Sign test with BINOM.DIST; rank sums with RANK.AVG

Equal variances, three or more groups (Levene, Bartlett)
In Excel: One-way ANOVA on absolute deviations from each group median

Post-hoc comparisons (Tukey)
In Excel: Fisher LSD with T.INV.2T; no studentized range in Excel

Unbalanced or multi-factor ANOVA and DOE analysis
In Excel: LINEST with coded columns for two-level factorials

Power and sample size
In Excel: Sample-size formula with NORM.S.INV

Process capability
In Excel: Cpk from AVERAGE and STDEV.S, or R-bar over d2

Control charts
In Excel: Limits from AVERAGE and constants, drawn as a line chart

Gage R&R
In Excel: ANOVA method built from DEVSQ

Weibull fitting
In Excel: LINEST on the linearized Weibull plot

Logistic or Poisson regression
In Excel: Solver maximizing the log-likelihood

PY(python_code, return_type) HELPER — scipy.stats and statsmodels for tests Excel lacks

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